



K S Tharunpranav 251801359-2025-AIDS ▾

K2

Started on	Monday, 24 August 2026, 9:03 PM
State	Finished
Completed on	Monday, 24 August 2026, 9:04 PM
Time taken	28 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int find_floor(int *arr, int low, int high, int x) {
5      if (low > high) {
6          return -1;
7      }
8
9      if (x >= arr[high]) {
10         return arr[high];
11     }
12
13     int mid = low + (high - low) / 2;
14
15     if (arr[mid] == x) {
16         return arr[mid];
17     }
18
19     if (mid > 0 && arr[mid - 1] <= x && x < arr[mid]) {
20         return arr[mid - 1];
21     }
22
23     if (x < arr[mid]) {
24         return find_floor(arr, low, mid - 1, x);
25     }
26
27     return find_floor(arr, mid + 1, high, x);
28 }
29
30 int main() {
31     int n;
32     if (scanf("%d", &n) != 1) return 0;
33
34     int *arr = (int *)malloc(n * sizeof(int));
35     if (arr == NULL) return 0;
36
37     for (int i = 0; i < n; i++) {
38         if (scanf("%d", &arr[i]) != 1) {
39             free(arr);
40             return 0;
41         }
42     }
43
44     int x;
45     if (scanf("%d", &x) != 1) {
46         free(arr);
47         return 0;
48     }
49
50     int floor_val = find_floor(arr, 0, n - 1, x);
51     printf("%d\n", floor_val);
52

```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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